

Constrained Realization: Following the Causal History of a Conscious Episode

Human-readable companion to Constrained Realization: A Cross-Level Framework for the Causal Provenance of Conscious Experience

Companion status. Human-readable companion to Constrained Realization: A Cross-Level Framework for the Causal Provenance of Conscious Experience. This version preserves the scientific claims and limits of the completed manuscript while presenting the motivating problem and argument in more direct language. It does not constitute a separate research contribution.

Abstract

A visible outcome does not tell us where its cause was constructed. A person may fail to report a stimulus because the stimulus was deprioritized, because a usable representation never formed, because represented content did not enter the neural processing regime associated with conscious perception, because that regime did not persist, or because memory or report failed afterward. Those possibilities can end in the same behavior while differing in causal history and, potentially, in phenomenal outcome. This companion presents Icarus as a framework for keeping those possibilities separate. It organizes established developmental, cognitive, cellular, circuit, and reporting processes into six explanatory roles: developmental construction, present constraint, candidate-state formation, an admission/recurrent realization complex, stabilization when separable, and downstream use. The proposal is not a new mechanism of consciousness and does not identify any dendritic, thalamocortical, molecular, developmental, or behavioral event with phenomenality. Its proposed contribution is a localization discipline: claims about one causal depth should be tested against neighboring alternatives, negative reports should be treated as underidentified until those alternatives are constrained, and historical causes should remain in the explanation only when they leave persistent internal or environmental changes that alter later processing. The framework is strongest for perceptual consciousness, where relevant cellular and circuit evidence is most developed. It remains deliberately agnostic about where phenomenality begins within the realization-related neural regime. If the six-role decomposition does not improve experimental attribution beyond established frameworks, it should collapse back into those components.

Keywords: consciousness; causal provenance; causal inference; developmental systems; neuroplasticity; apical amplification; reportability

1. The problem hiding behind an outcome

Suppose two people are shown the same faint stimulus and both say that they did not see it. The matching answers look simple, but they do not tell us that the same thing happened inside both systems. In one case, current goals or expectations may have pushed the stimulus below competing content. In another, a sufficiently strong content-bearing state may never have formed. In a third, the content may have been represented but failed to enter the neural regime associated with conscious perception. In a fourth, an early transition may have occurred but not persisted. In a fifth, experience may have occurred while later memory or the reporting pathway failed. The final answer - no report - is compatible with all of these causal histories.

That is the central problem addressed by Icarus. The framework does not begin by asking which single mechanism is consciousness. It asks a more limited question: when we observe a conscious report, a failed report, a discrimination, a remembered episode, or an action, what part of the causal history has the observation actually identified? The same endpoint can inherit differences that were constructed long before the moment of measurement, produced by the organism's present state, introduced during content formation,

introduced during realization-related neural processing, introduced during persistence, or introduced only afterward.

Consciousness makes this attribution problem unusually difficult because the measurement usually comes after the event we want to understand. A participant reports a stimulus, presses a button, recalls an image, or changes behavior. Those are indispensable observations, but the measurement channel is not automatically the phenomenon being measured. A later report can depend on processes that were not part of the target experience, and a missing report can fail for reasons that do not uniquely identify absence of experience (Aru et al., 2015). The methodological danger is not that reports are useless. It is that the endpoint can silently become a definition of everything that must have happened before it.

The broader warning is already established. Similar surface phenomena can arise from different underlying mechanisms, and explanatory claims should be sensitive to the level at which a difference is produced (Overgaard & Mogensen, 2011). Interventionist approaches to consciousness likewise distinguish causal difference-makers from correlations, enabling conditions, precursors, and downstream effects (Klein & Barron, 2020). Icarus does not claim either principle as new. It asks whether those principles become more operational if the causal trajectory of one candidate conscious episode is divided into a small set of roles that force neighboring explanations to be tested against one another.

The original human intuition behind the project can be stated without turning it into a scientific variable: when a person appears to choose, intend, notice, experience, remember, act, or fail to act, the visible endpoint does not reveal where the relevant difference was constructed. The mature paper reduces that intuition to a causal-provenance problem. 'Will' is not introduced as an entity, mechanism, or location. The scientifically defensible residue is the question of causal depth: how far back in the system's history, and at what functional stage, must we go before the observed difference becomes identifiable?

2. Before locating a cause, keep the target separate from its measurements

The framework begins with a conservative distinction between physical occurrence and phenomenal occurrence. A neural state can be physically present, decodable, causally active, or available to another process without the framework assuming that the state is itself phenomenal. Conversely, phenomenal occurrence is not defined here by later successful report, memory, voluntary control, or action. Presence is evidence of presence; access is evidence of access; use is evidence of use. None of those relations, by itself, establishes identity with experience.

This is a methodological seam rather than a metaphysical theory. It prevents a convenient readout from taking over the definition of the target. If neural information about a stimulus can be decoded after a person gives a negative report, that finding shows that information existed in some physical form. It does not by itself show that the information was consciously experienced. If a person later reconstructs details, that later reconstruction is also a causal episode with its own memory and reporting requirements. The reverse caution matters as well: failure of report does not uniquely identify failure of phenomenality. Bridging those claims requires evidence about what happened between content formation and the endpoint.

The same caution applies to biological mechanisms. Apical dendritic activity, layer-5 bursting, and recurrent thalamocortical dynamics are serious mechanistic candidates for processes associated with conscious perception, but a causal relation is not an identity relation. Evidence that mechanism Q changes detection, recurrent processing, or report supports a relation between Q and those outcomes under specified conditions. It does not automatically establish Q as phenomenality itself. Icarus therefore preserves the difference between causal-emergence or correlation claims and the much stronger claim that a mechanism is identical to experience.

The empirical scope is also narrower than the word consciousness can suggest. The strongest mechanistic evidence discussed here comes from perceptual tasks and, for direct cellular manipulation, largely from animal work. Human tests of circuit-level theories such as Apical Amplification Theory (AAT) and Dendritic Integration Theory (DIT) remain methodologically challenging (Dykstra et al., 2026). The framework can be proposed for conscious episodes more generally, but that generalization is theoretical. Perceptual consciousness is where the realization-side evidence is strongest.

With those boundaries in place, the causal history can be followed without pretending that every measurable step is already conscious. The purpose of the decomposition is precisely to leave phenomenal status open where the evidence leaves it open.

3. What established work already explains - and what remains to be organized

Most of the machinery needed by Icarus already exists in the literature. Developmental and learning theories explain how organism-specific capacities are constructed across time. Research on attention, motivation, affect, bodily state, expectation, and memory explains ways in which the current state of the organism changes what is prioritized. Consciousness science already distinguishes prerequisites, candidate neural correlates, and consequences of consciousness (Aru et al., 2015). AAT and DIT connect content-bearing activity to context-sensitive apical and apical-basal processing (Aru et al., 2020a; Marvan et al., 2021). Whyte et al. (2025) provide a burst-dependent thalamocortical model of perceptual awareness. None of those mechanisms becomes an Icarus discovery by being placed in one diagram.

Phillips (2023) is an especially important precedent because cooperative neuronal processing is discussed together with development, learning, attention, emotion, working memory, long-term memory, and conscious experience. Storm et al. (2024) likewise make clear that multiscale integration across consciousness theories is not itself novel. Cleeremans (2011) makes learning and plasticity central to the organization of conscious representations. Transactional and developmental-systems approaches already treat the person and context as reciprocally changing across time (Sameroff, 2010; Moore, 2026). The project therefore cannot claim novelty for developmental path dependence, plasticity, organism-environment recursion, context-sensitive processing, recurrent processing, or cross-scale integration.

What remains is organizational and inferential. Icarus asks whether a six-role causal map provides more precise experimental bookkeeping than these components provide separately. The map is coupled to four additional disciplines: a negative-report matrix that makes several hidden causes explicit; a measurement seam that keeps phenomenality distinct from representation, accessibility, memory, report, and action; a requirement to collect evidence at neighboring causal depths before localizing a difference; and a longitudinal rule that keeps historical causes only when they leave a persistent update that changes later processing.

This residual contribution is intentionally modest. Generic many-to-one causation and interventionism are inputs, not discoveries. Developmental history and plasticity are inputs, not discoveries. Candidate-to-realization biology is inherited from existing cellular and circuit theories. The framework earns a place only if arranging these precedents by causal provenance changes what an experiment measures, manipulates, compares, or is allowed to conclude.

Table 1. Nearest relevant precedents and the residual Icarus integration

Framework / precedent	What it already establishes	What it does not by itself establish	Relation to Icarus
Aru et al. (2015)	Prerequisite/NCC/consequence separation; downstream contamination problem	The six-role construction-to-use localization matrix	Provides the broad methodological separation

Framework / precedent	What it already establishes	What it does not by itself establish	Relation to Icarus
Overgaard & Mogensen (2011)	Similar surface cognitive/conscious outcomes may arise from dissimilar mechanisms; levels matter	The Icarus six-role negative-report matrix and longitudinal update rule	Establishes the generic surface-equivalence warning
Klein & Barron (2020)	Interventionist causal explanation of consciousness; background/enabling/downstream distinctions	The Icarus role taxonomy and its specific evidentiary matrix	Establishes intervention-based causal localization in principle
AAT / DIT / Whyte et al. (2025)	Context-sensitive candidate processing, dendritic integration, recurrent thalamocortical awareness models	Developmental-to-downstream provenance ordering or phenomenality identity	Provide realization-side mechanistic anchors
Phillips (2023) / Storm et al. (2024)	Development, contextual neuronal cooperation, learning/memory, conscious experience, multiscale integration	The same six-role localization matrix and longitudinal collapse rule	Broad mechanistic and multiscale precedents
Cleeremans (2011)	Learning/plasticity as central to the organization of conscious representation	The Icarus causal-role matrix and negative-report localization scheme	Owens generic plasticity-consciousness coupling
Moore (2026) / Sameroff (2010)	Historical developmental systems; reciprocal person-context change; experience-sensitive biological development	A consciousness-specific candidate/realization/report matrix	Own broad recursive developmental framing

4. Six different questions can sit behind one conscious episode

The six roles are easiest to understand as six different questions rather than six anatomical boxes. A single structure may contribute to more than one role, several roles may overlap in time, and causal influence can loop back to change later episodes. The labels are retained because each question supports a different kind of evidence and a different failure mode.

Construction asks: why does this organism have this particular space of possible states and transitions? Brain development is not a finished genetic blueprint being read out into an adult system. Prenatal conditions, bodily state, spontaneous activity, sensory and social input, gene regulation, learning, and the organism's own behavior all participate in a historically extended developmental system (Stiles & Jernigan, 2010; Sameroff, 2010; Moore, 2026). DNA sequence contributes constraints on developmental possibilities, but experience-sensitive transcriptional and epigenetic states can change. Construction therefore does not explain phenomenality. It explains why later processing occurs in this particular architecture, with this connectivity, tuning, plasticity, learned organization, and repertoire of physically available trajectories.

Present constraint asks: given the system that exists now, what makes some possibilities more influential than others at this moment? Current bodily state, arousal, goals, learned expectation, emotion, environmental context, recently active representations, task demands, and memory-dependent information can alter probabilities, gain, or transition accessibility without being identical to the candidate content. AAT provides a cellular precedent for context-sensitive modulation: apical processing can reflect needs, goals, emotion, and other contextual information that affects whether local content reaches a stronger processing regime (Marvan et al., 2021). The provenance point is that even a 'current' signal has a history. A goal was learned or adopted;

an expectation reflects prior experience; a memory signal depends on storage and retrieval; a bodily state reflects ongoing regulation.

Candidate-state formation asks: is the relevant content physically present in a form that can participate in further competition, amplification, recurrence, or use? A candidate need not be a single localized representation. It may be distributed, partial, competitive, recurrent, or internally generated. The key is that content-bearing neural organization can be measurable before phenomenal occurrence has been established. This distinction is strongly preceded by work showing processing without ordinary awareness and by AAT's separation between content and the contextual amplification proposed to alter conscious status (Marvan et al., 2021). The practical value is straightforward: if the candidate was never present, a later failure cannot be localized specifically to the mechanisms proposed to realize it.

The admission/recurrent realization complex asks: what changes when a candidate possibility becomes an actually instantiated neural regime associated with conscious perception? Here the paper uses established cellular and circuit work rather than inventing a new transition. Layer-5 pyramidal neurons can integrate inputs arriving at different compartments through nonlinear dendritic electrogenesis (Larkum et al., 1999). Active cortical dendrites can causally modulate perceptual detection, and dendritic currents can gate descending cortical outputs during tactile perception (Takahashi et al., 2016, 2020). General anesthesia can disrupt coupling between apical and somatic compartments while leaving some feed-forward responsiveness comparatively intact (Suzuki & Larkum, 2020). AAT and DIT interpret related apical and apical-basal processes in consciousness-relevant terms (Aru et al., 2020a; Marvan et al., 2021), while Whyte et al. (2025) model burst-dependent thalamocortical dynamics associated with perceptual awareness.

The phrase realization complex is deliberately noncommittal. It can include apical-basal interaction, regenerative dendritic activity, layer-5 burst output, higher-order thalamic recruitment, and recurrent cortical-thalamic dynamics. It says that a physical transition occurred; it does not say why that transition is phenomenal or where within that family phenomenality begins. A local nonlinear event could be part of an early transition, could initiate a larger recurrent process, or could be inseparable from a coupled recurrent regime. Current evidence does not justify the identity claim that a dendritic calcium event, a burst, or a recurrent loop simply is consciousness.

Stabilization asks a narrower empirical question: if an early realization-related transition occurs, can the resulting regime fail to persist? Conceptually, initial admission and later persistence are different questions. Empirically, they may turn out to be the same process. Stabilization therefore remains only provisionally distinct. If an intervention can preserve an early transition while selectively reducing duration, dominance, or resistance to replacement, the distinction gains support. If no such dissociation can be found, stabilization should collapse into realization rather than survive because a diagram has two boxes.

Downstream use asks what happens after, or because of, the target state: report, memory formation or retrieval, deliberate decision, and overt action. These processes can feed back into ongoing perception, so they are not assumed to be cleanly post-phenomenal in every task. The category exists to prevent the measurement channel from being mistaken for the target. A person can fail at a downstream step even when upstream content and realization-related dynamics are comparatively preserved. Conversely, successful behavior can be produced through different upstream mixtures of evidence, priors, candidate dynamics, persistence, and decision processes.

The roles therefore form a causal ordering without requiring a fixed serial anatomy: developmental construction -> present constraint -> candidate formation -> realization-related processing -> stabilization when separable -> downstream use. The ordering matters because an alteration at one depth changes what later observations can mean.

5. One negative report can hide several different failures

The negative-report case makes the framework concrete. Let $R(X)$ mean that a participant successfully reports target content X . Observing not- $R(X)$ does not tell us which hidden cause occurred. Current constraints may have favored something else; a sufficiently strong candidate representation may not have formed; a candidate may have formed but realization-related processing may have been impaired; an early transition may have occurred without persistence; or memory/report mechanisms may have failed.

These alternatives are useful only if they imply different evidence. A candidate-level failure predicts weak or absent content-bearing representation. A realization-level failure predicts measurable candidate content together with altered realization-related dynamics. A stabilization-level failure, if the distinction is real, predicts evidence of an early transition with reduced persistence. A downstream-use failure predicts comparatively preserved upstream evidence with impaired behavioral expression or later access. The same final behavior therefore leads to different next measurements depending on where the difference is hypothesized to lie.

This does not prove that experience was present in any particular negative-report trial. The matrix is a localization aid, not a way to infer hidden phenomenality by definition. Its value is narrower: it blocks the shortcut from one terminal observation to one preferred causal story. Behavioral equivalence does not establish mechanistic equivalence, and mechanistic nonequivalence prevents automatic phenomenal equivalence.

The same warning applies to positive responses. Two people can give the same correct answer while relying on different sensory evidence, expectations, candidate dynamics, recurrence, persistence, or decision processes. Agreement at the endpoint is therefore compatible with differences earlier in the trajectory.

Table 2. Causal-dissociation matrix: one negative-report endpoint, multiple causal depths

Altered causal depth	Candidate-content evidence	Realization-related evidence	Persistence	Report	Interpretive consequence
Current constraint	Changed or deprioritized	Competent	Competent	Changed/absent	Different content favored; no localization to realization
Candidate formation	Absent or altered	Competent in principle	Competent in principle	Changed/absent	Content may never reach the relevant representation
Realization complex	Present	Impaired	Often impaired	Changed/absent	Candidate can exist without entering the target processing regime
Stabilization (if separable)	Initially present	Early transition may occur	Impaired	Weak/absent	Tests whether admission and persistence can dissociate
Downstream use	Present	Present	Present	Impaired	Endpoint does not identify whether phenomenality was absent

6. What would count as locating the difference

Interventionist causal reasoning is not new to Icarus (Klein & Barron, 2020). The framework adds value only if the six roles force more discriminating designs. The practical rule is to avoid making a claim about one

depth from evidence collected only at the endpoint. Whenever possible, measure or manipulate at least two neighboring depths.

For construction and historical update, the signature is not a momentary loss of consciousness but a durable change in the repertoire, weighting, or susceptibility of later processing. Developmental perturbations, learned changes, or bounded interventions can be relevant if they leave persistent molecular, synaptic, structural, representational, or environmental changes and those changes can be linked to later processing. Doron et al. (2020), for example, show a learning-related role for perirhinal input to neocortical layer 1 together with changes relevant to dendritic excitability. The historical claim becomes stronger when the initiating event, persistent update, and altered later episode can all be connected.

For present constraint, the intervention should alter context, expectation, goal, bodily state, or related signals while leaving candidate-generation and realization capacity otherwise available. The expected result is preferentially a change in which content is favored rather than a general abolition of conscious processing. This prediction is preceded by contextual accounts; Icarus uses it to localize what such a manipulation has actually identified.

For candidate formation, the crucial evidence is whether target content exists in a measurable neural form. Candidate measurements are therefore a necessary control for claims about realization. If the content was absent before the realization-related manipulation could matter, the failed endpoint does not isolate realization.

For realization, the stronger dissociation preserves measurable candidate content while selectively altering the cellular or circuit events associated with conscious perception. The dendritic, anesthetic, and thalamocortical literatures provide important precedents for this logic (Takahashi et al., 2016, 2020; Suzuki & Larkum, 2020; Aru et al., 2020a; Marvan et al., 2021; Whyte et al., 2025). But a behavioral or physiological dissociation still does not settle phenomenality without additional assumptions and measurements.

For stabilization, the experiment must separate entry into the relevant regime from its persistence. If the early transition cannot be preserved while later duration or dominance is selectively changed, stabilization should not remain a separate causal stage.

For downstream use, memory, reporting, or action should be perturbed while upstream candidate and realization-related evidence is measured. Interpretation is difficult because evidence of preserved experience often relies on those same downstream channels. The appropriate result may therefore be underidentification rather than a definitive statement that phenomenality survived.

Across all roles, adjacent-depth evidence is a discipline rather than a magic criterion. Measuring a candidate while perturbing realization can rule out some alternatives; measuring early realization markers while perturbing recurrence can rule out others; measuring upstream dynamics while perturbing report can constrain still others. Strong causal attribution still depends on valid measures, selective interventions where feasible, and control of competing pathways. A terminal behavior alone is not enough.

7. History can change the next episode - but history must earn explanatory status

The deepest part of the framework is historical, but it is also the part most likely to become vague if left unconstrained. Developmental systems, learning-dependent plasticity, and reciprocal person-context change are already established (Cleeremans, 2011; Sameroff, 2010; Moore, 2026). Icarus uses those ideas for a narrower question: when does an earlier event remain part of the causal explanation of a later conscious episode?

Let C_t denote the organism's effective constraint architecture at time t , X_t the relevant bodily and environmental conditions, and E_t an event processed under those conditions. Some events leave persistent internal changes - learning, molecular regulation, synaptic or structural plasticity, altered learned models -

represented schematically as ΔC_t . Downstream action can also change later exposure or environmental conditions. The qualitative loop is: $(C_t, X_t) \rightarrow E_t \rightarrow \{\Delta C_t, \text{action}\} \rightarrow (C_{t+1}, X_{t+1})$. This loop is not proposed as novel. The added rule is that history remains scientifically relevant only if the update can be connected to changed later processing.

Molecular examples show why 'history' need not mean a literal stored narrative. The DNA sequence is comparatively stable, but methylation, demethylation, chromatin state, transcription, and related regulatory processes can respond to experience. In adult rats, contextual fear learning altered hippocampal DNA-methyltransferase expression and methylation of memory-related genes, and inhibition of DNA methyltransferases disrupted memory formation (Miller & Sweatt, 2007). Maternal-care differences in rats were associated with persistent methylation and expression differences at a hippocampal glucocorticoid-receptor promoter, with cross-fostering altering the developmental outcome (Weaver et al., 2004). Human postmortem hippocampal findings associate childhood-abuse history with altered NR3C1 promoter methylation and glucocorticoid-receptor expression, while remaining observational rather than a direct causal chain (McGowan et al., 2009). These findings do not make DNA a selector of conscious content. They show that prior conditions can become physically embedded in systems that later shape susceptibility.

The hippocampus is relevant in this discussion as a history-sensitive contextual system, not as a mandatory gateway for all conscious input. Stress can produce dentate-gyrus-specific molecular changes, with methyl-donor manipulations altering later molecular and behavioral responses (Saunders et al., 2016). Environmental enrichment provides a different example: genetically homogeneous mice in the same enriched environment developed persistent individualized behavior accompanied by differences in hippocampal neurogenesis and DNA methylation (Zocher et al., 2020). The point is not that one molecular mark determines a later experience. It is that an architecture carrying prior history remains physically editable.

Rapid-plasticity interventions make the timescale distinction especially visible. In mice, a single psilocybin exposure increased formation and size of dendritic spines on layer-5 pyramidal neurons, with a subset of changes persisting for weeks (Shao et al., 2021). A single exposure to the serotonergic psychedelic DOI produced longer-lived synaptic and epigenomic changes together with altered fear-extinction behavior (de la Fuente Revenga et al., 2021). In humans, dense longitudinal neuroimaging after psilocybin showed large acute changes in network organization and more persistent changes in some connectivity measures over subsequent weeks (Siegel et al., 2024). These results do not show that the subjective psychedelic state caused the later changes and do not identify a mechanism of phenomenality. They demonstrate the narrower point that an acute perturbation can be followed by changes that outlast the perturbation and may alter future processing conditions.

Similar starting conditions also do not guarantee identical trajectories. Monozygotic twins can diverge epigenetically across life (Fraga et al., 2005). Prospective twin work has associated differential stress histories with later within-pair personality differences while remaining observational and limited in sample size (Torgersen & Janson, 2002). Genetically identical mice in shared enriched environments can likewise develop persistent individual differences (Zocher et al., 2020). These examples support historical divergence; they do not provide a consciousness-specific mechanism.

Learned models provide the same logic at a psychological level. Long-term priors can alter visual perception through measurable long-range feedback (Hardstone et al., 2021). A belief, expectation, goal, or learned model can matter causally when it is physically represented and changes attention, valuation, emotion, action, or environmental exposure. The framework does not require a nonphysical belief to act on neurons. It requires a physically implemented learned relation to alter later processing or behavior. Action can then change the situations and evidence encountered later, adding an environmental feedback path.

The resulting view is one of bounded change, not unlimited self-rewriting. Anatomy, connectivity, receptor distributions, age, developmental history, physiology, pathology, environmental access, prior learning, and the

circuits recruited by a new event constrain what can change and how much. Plasticity means that history can matter; it does not mean that every disposition can be deliberately rewritten at will.

Most importantly, historical explanation has a collapse condition. An earlier event earns a place in the later provenance analysis only when evidence links it to a persistent internal or environmental update and links that update to altered later susceptibility, weighting, candidate formation, realization-related processing, or relevant behavior. If contemporaneous state variables fully capture the susceptibility of interest, the historical path adds genealogy but no independent explanatory level. Provenance is therefore not permission to tell an indefinitely long causal story. It is a claim that causal depth matters only where removing that depth changes what can be predicted or explained.

$$(C_t, X_t) \rightarrow E_t \rightarrow \{\Delta C_t, \text{action}\} \rightarrow (C_{t+1}, X_{t+1})$$

Figure 1. Recursive provenance and organism-environment coupling. Events processed under the current constraint architecture can update that architecture through learning and plasticity, while downstream action can alter later exposure or environmental conditions. Both pathways can shape later processing; phenomenal occurrence is not assumed to be necessary for either update. Arrows denote possible causal classes rather than universal transitions.

8. Content can come from outside or inside; realization is a separate question

A model of conscious realization cannot assume that conscious content always begins with current sensory input. Dreams, imagery, and other internally generated experiences make the source of content a separate question from whether content is realized. An externally anchored candidate may depend heavily on current sensory input; an internally generated candidate may depend more on memory, association, spontaneous dynamics, imagery, or bodily signals. Either could, in principle, converge on overlapping realization machinery.

Aru et al. (2020b) proposed 'apical drive' as a possible cellular mechanism relevant to dreaming, extending apical-function ideas beyond amplification of externally driven input. The proposal is not established as the mechanism of dreaming, but it illustrates the distinction: the same general cellular architecture can be considered when internally generated activity supplies more of the effective drive.

This matters for interpretation. Reduced connectedness to the external environment does not by itself establish loss of phenomenal experience. Conversely, evidence from perceptual detection cannot simply be generalized to all conscious thought, emotion, dreaming, pain, or self-experience. The source of content, the formation of a candidate, and the realization of that candidate are related but not identical causal questions.

The phrase constrained realization compresses the full architecture. A biological system occupies a physical state within a historically constructed space of accessible transitions. Development, learning, plasticity, and current context change which trajectories are feasible or favored. A candidate state is an actual neural configuration within that space. Realization refers to a further physical transition into a regime associated with conscious processing. Constraint therefore addresses why some trajectories are more available or influential than others; realization addresses what physically occurs rather than what was merely possible.

The state-space language is intentionally qualitative. The paper does not provide a measured high-dimensional manifold, an energy landscape, a neural code, or a solved transition equation. Cellular events such as dendritic electrogenesis are useful because they are mechanistically grounded examples of physical state transition, not because physical reality entails phenomenal identity. 'Consciousness = constrained realization' should therefore be read as a theoretical summary of the decomposition, not as a solved equation or an identity claim.

9. When the framework should shrink, collapse, or disappear

A provenance framework can become unfalsifiable if every distinction is protected simply because it sounds plausible. Icarus therefore includes explicit conditions under which parts of the architecture should be removed.

Admission and stabilization should collapse into a single realization process if they cannot be independently measured or manipulated. Candidate and realization should collapse if no content-bearing candidate can be identified independently of the dynamics proposed to realize it. A distinction that never supports a different observation, intervention, prediction, or failure condition is not earning its place.

Historical provenance should also disappear from a particular analysis if present-state variables fully capture the relevant susceptibility. Persistent molecular or structural change is not enough merely because it persists; it must alter later weighting, transition, candidate formation, realization-related processing, or behavior relevant to the claim. Action-mediated environmental feedback likewise adds nothing distinctive if established ecological, learning, transactional, or developmental-systems models already capture everything needed for the target analysis.

The entire framework is subsumed if an established theory already contains the same six-role construction-to-constraint-to-candidate-to-realization-to-stabilization-to-use decomposition together with an equivalent negative-report matrix, phenomenality/report seam, adjacent-depth evidence discipline, and longitudinal update-to-later-processing rule. The literature contains close partial precedents across these elements - including Aru et al. (2015), Overgaard and Mogensen (2011), Cleeremans (2011), AAT/DIT, Klein and Barron (2020), Phillips (2023), Storm et al. (2024), Whyte et al. (2025), and Moore (2026) - but the completed manuscript's targeted audit did not establish one source or tightly connected framework containing the complete integration.

Most importantly, Icarus fails if causal depth does not change experimental interpretation. A diagram that merely sorts known mechanisms is not a substantive contribution. The framework is useful only if knowing the candidate depth changes which measurement is needed, which alternative must be ruled out, or what a result can support.

Several empirical limitations remain. The strongest realization-side evidence concerns perceptual detection and awareness. Direct causal work on apical dendrites is largely from animal models, and animal detection behavior cannot by itself adjudicate phenomenality. Human circuit-level tests remain incomplete (Dykstra et al., 2026). The recursive-provenance evidence is heterogeneous: causal molecular and structural demonstrations are strongest in animal models; human epigenetic work is often observational; and human psychedelic studies do not currently supply a before-and-after neural epigenomic chain linking subjective experience to later conscious processing. Peripheral molecular changes would not establish neural modification.

The six roles are functional explanatory positions, not guaranteed anatomical modules. One neural structure can contribute to several roles, and several roles can overlap in time. The state-space description is qualitative. Plasticity is bounded. No claim is made that personality, preference, or goals can be deliberately rewritten without constraint. The framework also does not solve the explanatory gap between neural dynamics and phenomenality. It leaves open whether phenomenality begins with a local nonlinear event, a recurrent circuit regime, a broader organization, or a mechanism not considered here.

The novelty claim is therefore synthetic and compositional. Component mechanisms and broad methodological principles are established. The framework should be judged only on whether its specific localization architecture, evidence matrix, measurement seam, and longitudinal collapse rule improve experimental design and interpretation.

10. Conclusion

A conscious episode is not an isolated endpoint. It occurs in a system whose architecture was built across development, whose present state changes what is favored, whose content must be physically instantiated, whose candidate activity may or may not enter and persist within a realization-related neural regime, and whose later report, memory, decision, or action adds further causal steps. Some episodes and actions then feed back into the system through learning, plasticity, altered models, and changed environments, affecting later episodes.

None of those components is proposed here as a newly discovered consciousness mechanism. The general fact that one endpoint can conceal multiple mechanisms is also not new. Icarus makes that fact concrete for a candidate conscious episode by placing six causal roles on the same trajectory and asking what evidence would distinguish neighboring alternatives.

The practical lesson is simple even when the biology is not: do not ask a terminal observation to tell you more causal history than it contains. A failed report may have been constructed before content formed, during realization-related processing, during persistence, or only during later memory and response. A successful report can likewise be reached through different upstream paths. Causal attribution improves when experiments measure or manipulate more than one depth and preserve the distinction between phenomenality and the channels used to infer it.

Longitudinally, history remains part of the explanation only when it leaves a persistent internal or environmental change that alters later processing. If present state fully captures the relevant susceptibility, the historical path should be dropped. Mechanistically, realization-related biology remains a causal candidate rather than an identity claim. Where phenomenality begins remains unresolved.

Constrained realization is therefore a bounded research framework: historically constructed constraints shape which trajectories are available and favored; realization names the physical transition from possibility to an instantiated neural regime associated with conscious processing. The framework succeeds only if causal depth changes what experiments mean. If it does not, it should collapse back into the established components from which it was assembled.

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